

Avian Speciation

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What is a species?

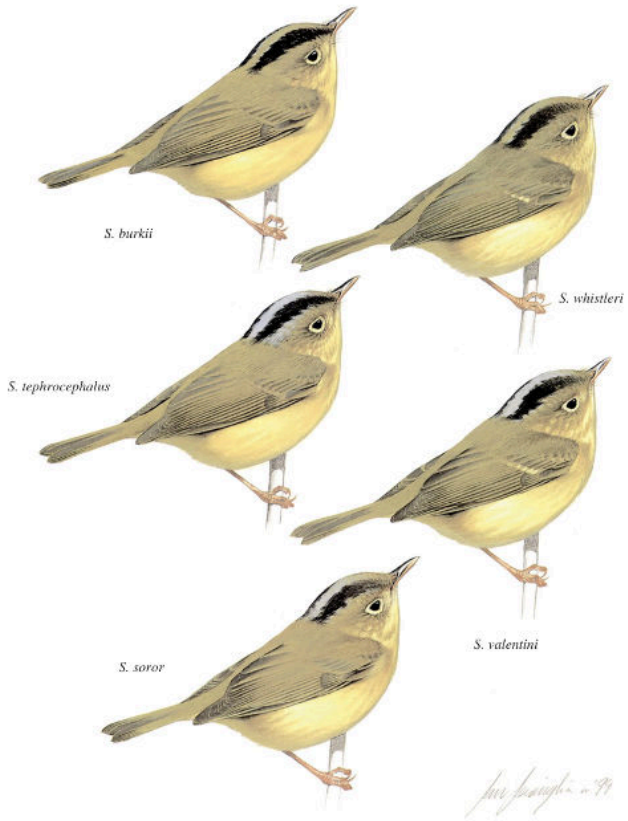




(*Empidonax difficilis* / *Empidonax occidentalis*)

Species concepts

- 24 species concepts and counting (Mayden 1997)
- Widespread confusion between conceptual basis and operational criteria (*definitions* versus *delimitation*)
- Conflict from alternate definitions, insufficient data...



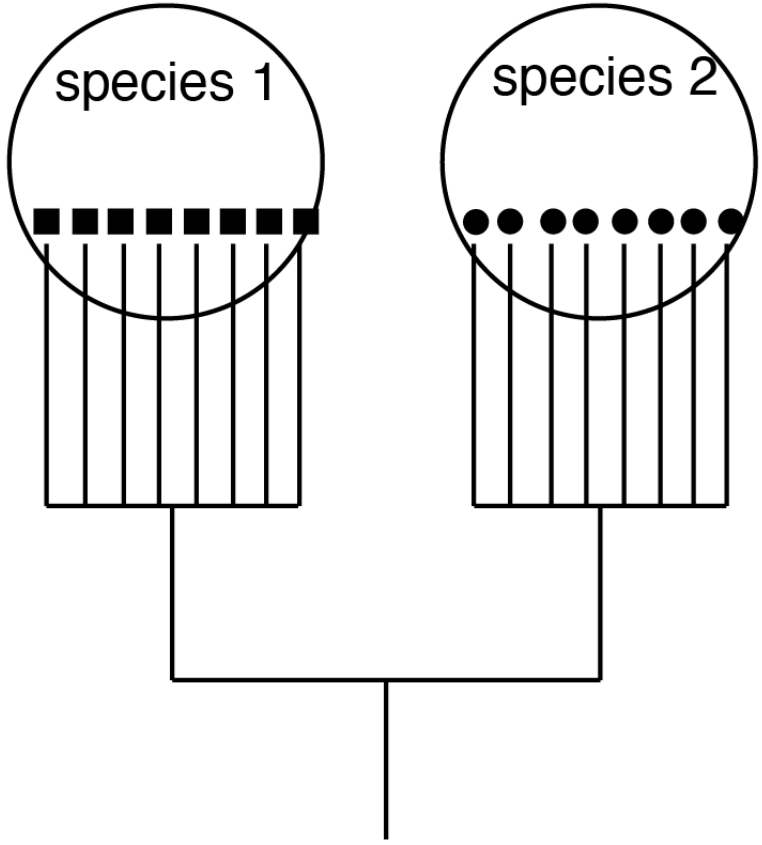
Species concepts

Species concept	Property(ies)	Advocates/references
Biological	Interbreeding (natural reproduction resulting in viable and fertile offspring)	Wright (1940); Mayr (1942); Dobzhansky (1950)
Isolation	*Intrinsic reproductive isolation (absence of interbreeding between heterospecific organisms based on intrinsic properties, as opposed to extrinsic [geographic] barriers)	Mayr (1942); Dobzhansky (1970)
Recognition	*Shared specific mate recognition or fertilization system (mechanisms by which conspecific organisms, or their gametes, recognize one another for mating and fertilization)	Paterson (1985); Masters et al. (1987); Lambert and Spencer (1995)
Ecological	*Same niche or adaptive zone (all components of the environment with which conspecific organisms interact)	Van Valen (1976); Andersson (1990)
Evolutionary	Unique evolutionary role, tendencies, and historical fate	Simpson (1951); Wiley (1978); Mayden (1997)
(some interpretations)	*Diagnosability (qualitative, fixed difference)	Grismer (1999, 2001)
Cohesion	Phenotypic cohesion (genetic or demographic exchangeability)	Templeton (1989, 1998a)
Phylogenetic	Heterogeneous (see next four entries)	(see next four entries)
Hennigian	Ancestor becomes extinct when lineage splits	Hennig (1966); Ridley (1989); Meier and Willmann (2000)
Monophyletic	*Monophyly (consisting of an ancestor and all of its descendants; commonly inferred from possession of shared derived character states)	Rosen (1979); Donoghue (1985); Mishler (1985)
Genealogical	*Exclusive coalescence of alleles (all alleles of a given gene are descended from a common ancestral allele not shared with those of other species)	Baum and Shaw (1995); see also Avise and Ball (1990)
Diagnosable	*Diagnosability (qualitative, fixed difference)	Nelson and Platnick (1981); Cracraft (1983); Nixon and Wheeler (1990)
Phenetic	*Form a phenetic cluster (quantitative difference)	Michener (1970); Sokal and Crovello (1970); Sneath and Sokal (1973)
Genotypic cluster (definition)	*Form a genotypic cluster (deficits of genetic intermediates; e.g., heterozygotes)	Mallet (1995)

(De Queiroz 2007)

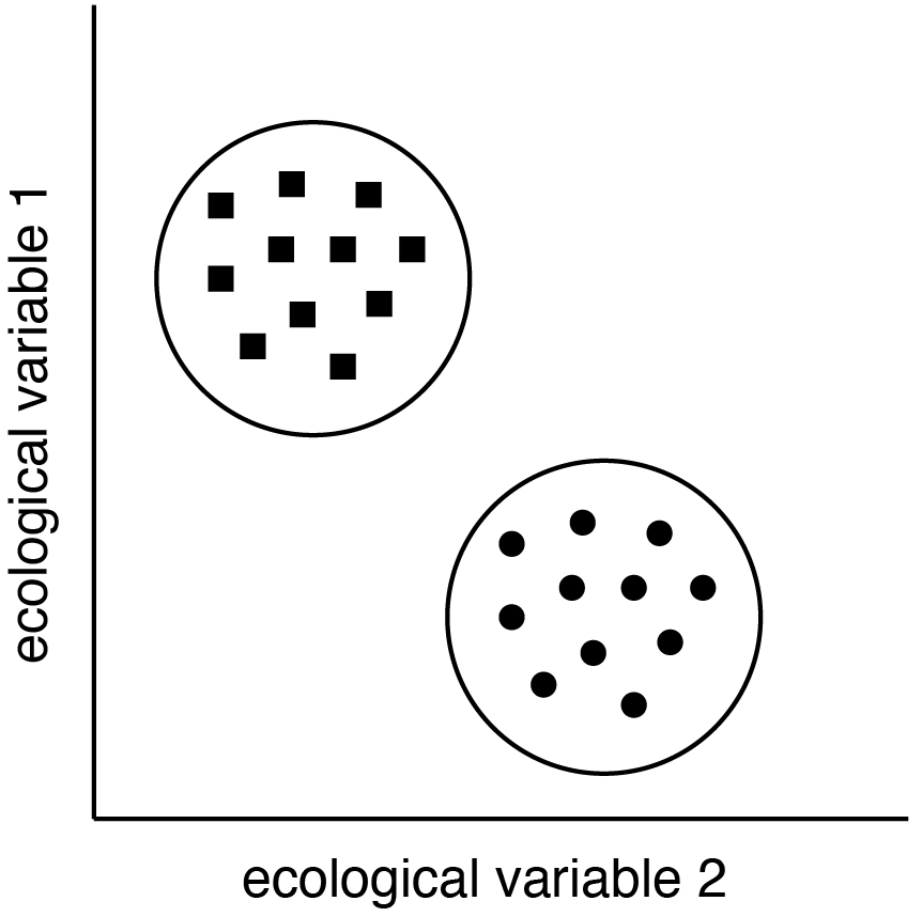
Species concepts

- *Phylogenetic species concept*: A species is a group of individuals that conform to one or more phylogenetic criteria (reciprocal monophyly, coalescence of all gene trees)



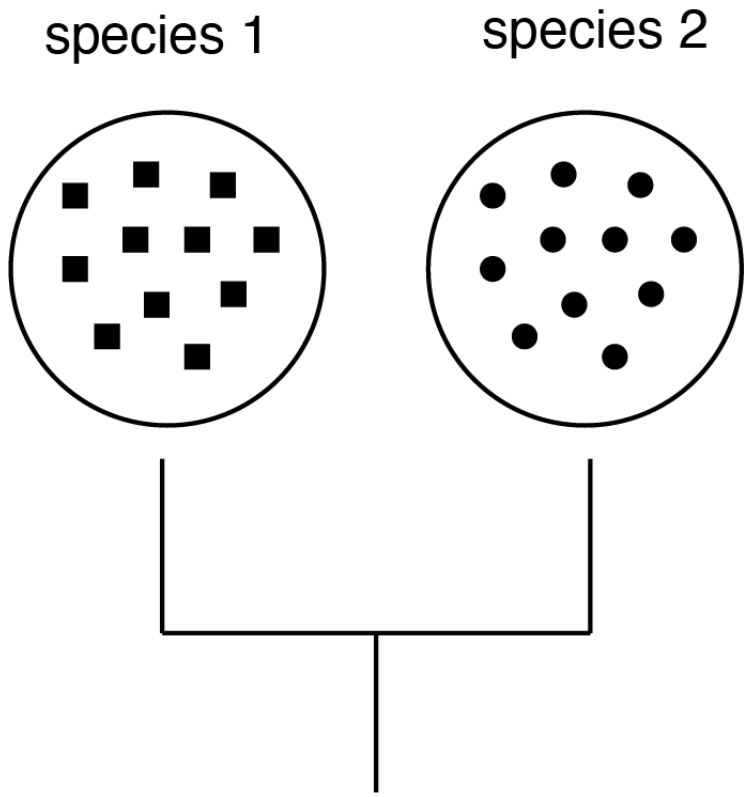
Species concepts

- *Ecological species concept*: A species is a group of individuals that share the same ecological niche or adaptive zone



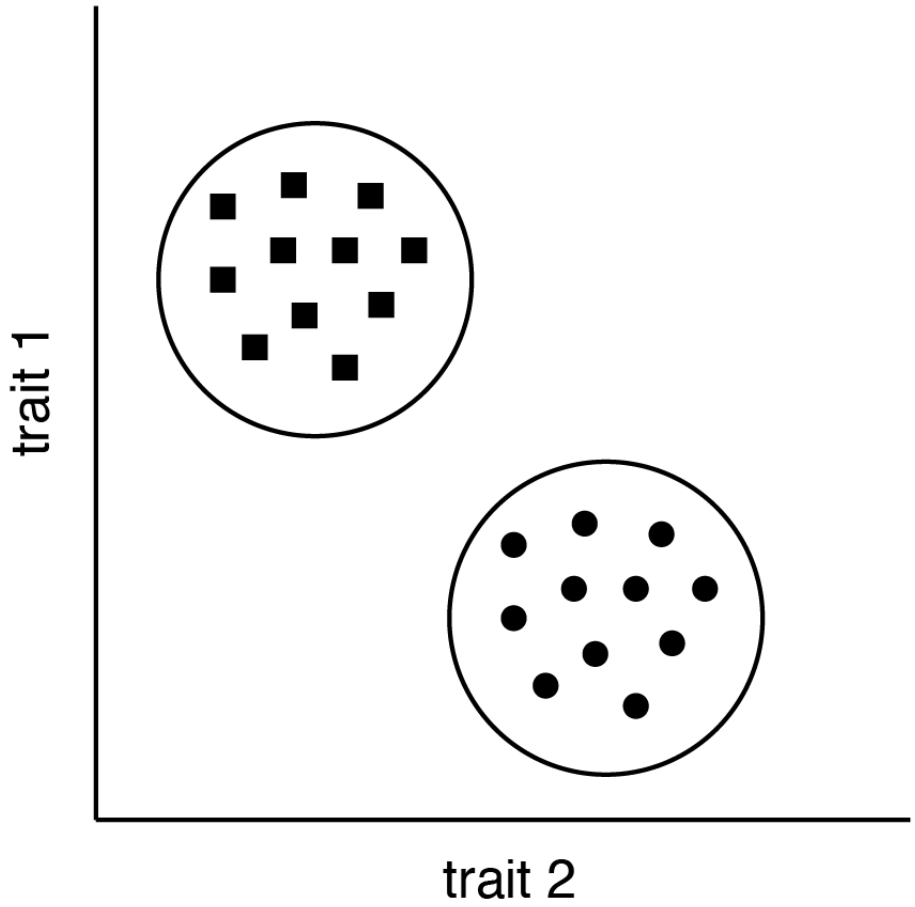
Species concepts

- *Evolutionary species concept*: A species is a group of individuals with the same evolutionary role, tendencies, and historical fate



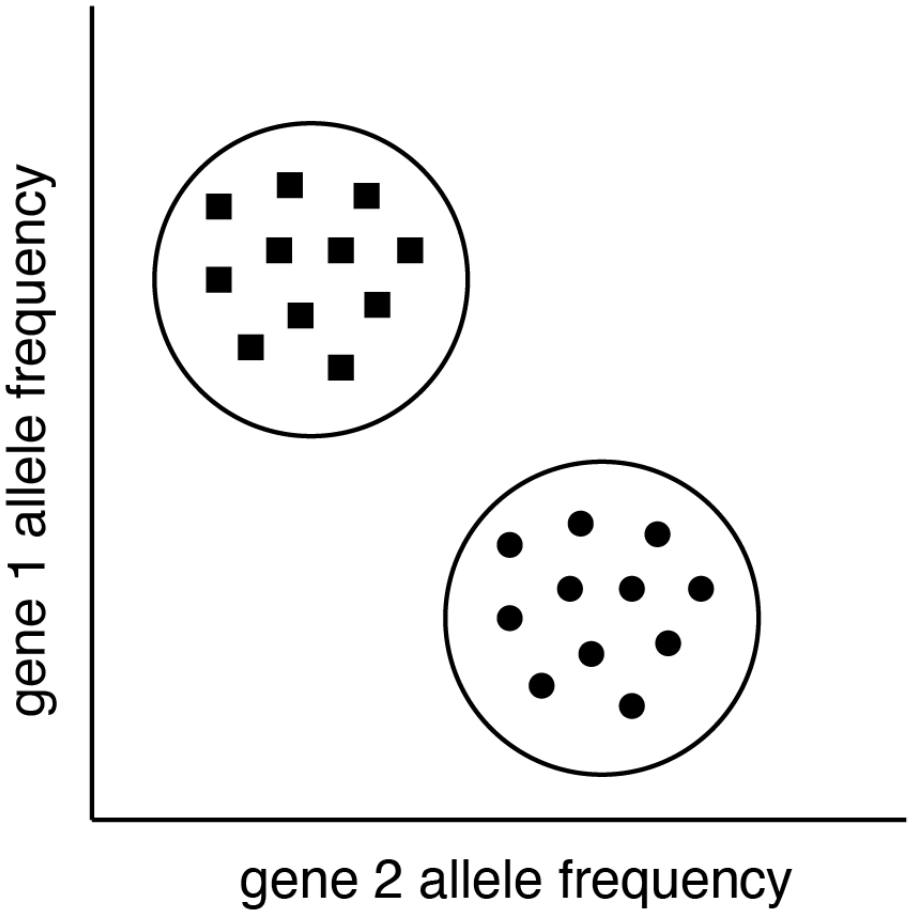
Species concepts

- *Phenetic cluster species concept*: A species is a group of individuals that form a cluster in morphological space separated from other such groups (similar trait values for multiple features)



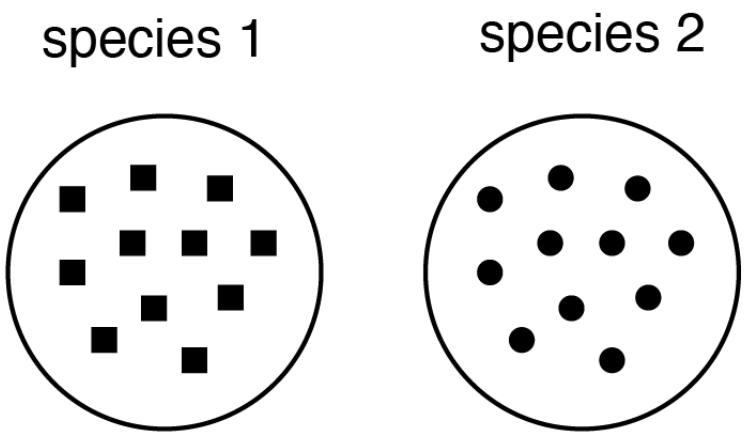
Species concepts

- *Genotypic cluster species concept*: A species is a group of individuals that form a cluster in genoptypic space separated from other such groups (similar allele frequencies at multiple loci)



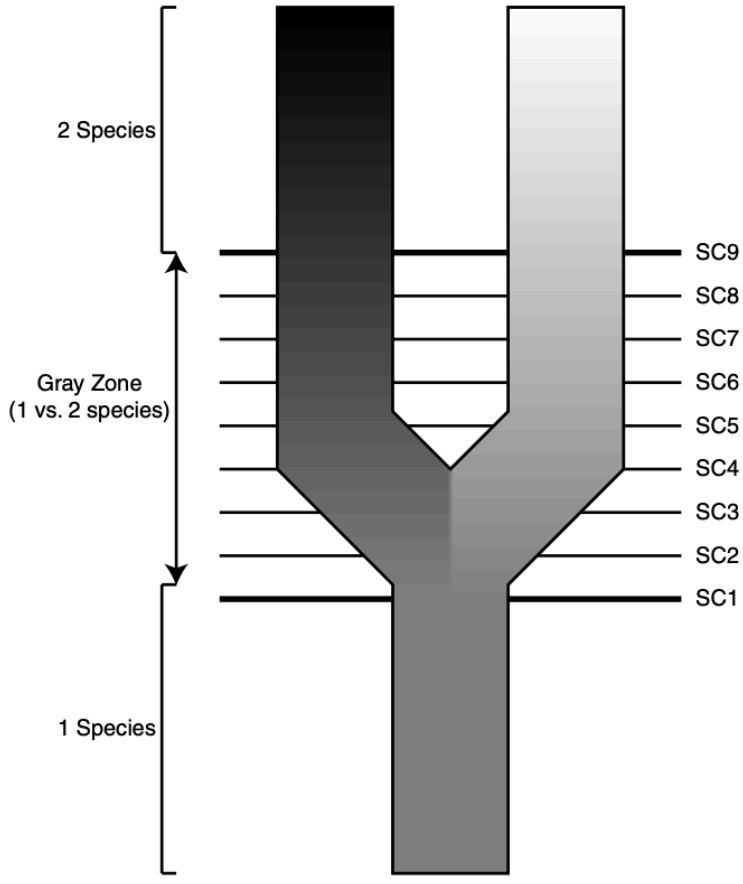
Species concepts

- *Biological species concept*: A species is a group of actively or potentially interbeeding individuals separated from other such groups by intrinsic or extrinsic reproductive isolation (not only geography)



Speciation is a spectrum

- Another source of conflict: some taxa caught in middle of the process!



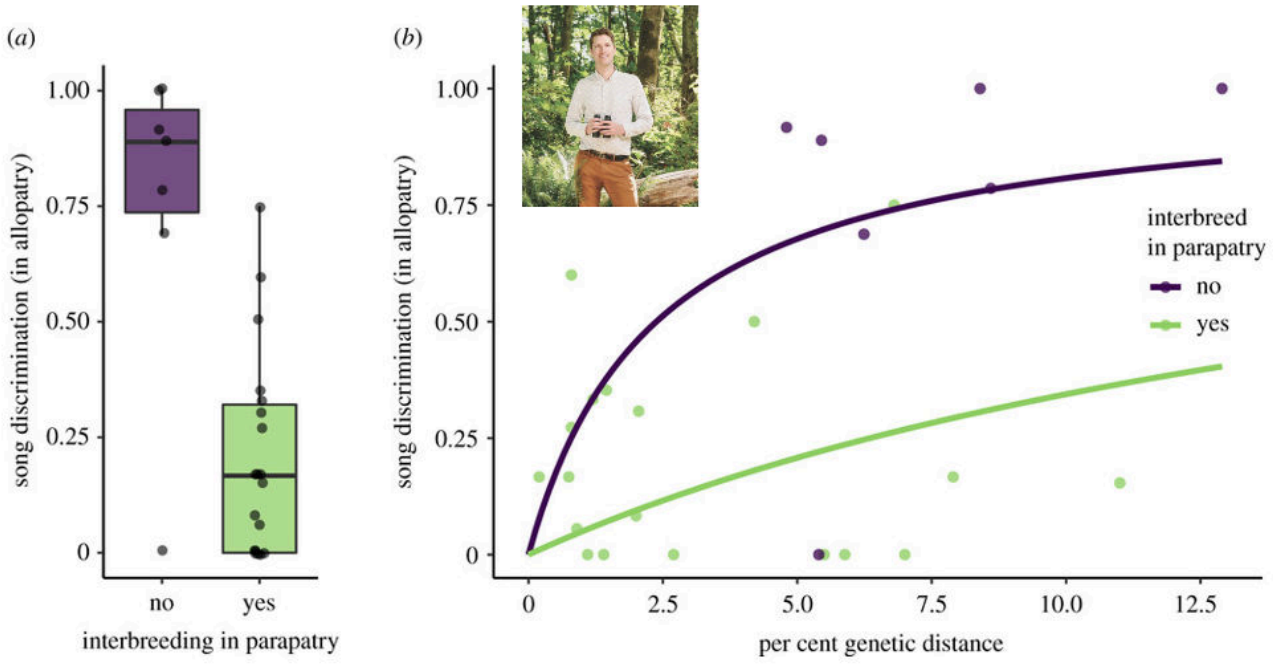
(De Queiroz 2007)

Which of the following is a *criterion* for delimiting species?

a) species have genomes that are greater than 5% different than their closest relative
b) species are groups of individuals that share an ecological niche
c) species are members of a lineage with a shared evolutionary trajectory
d) none of the above
e) all of the above

Prezygotic Isolation

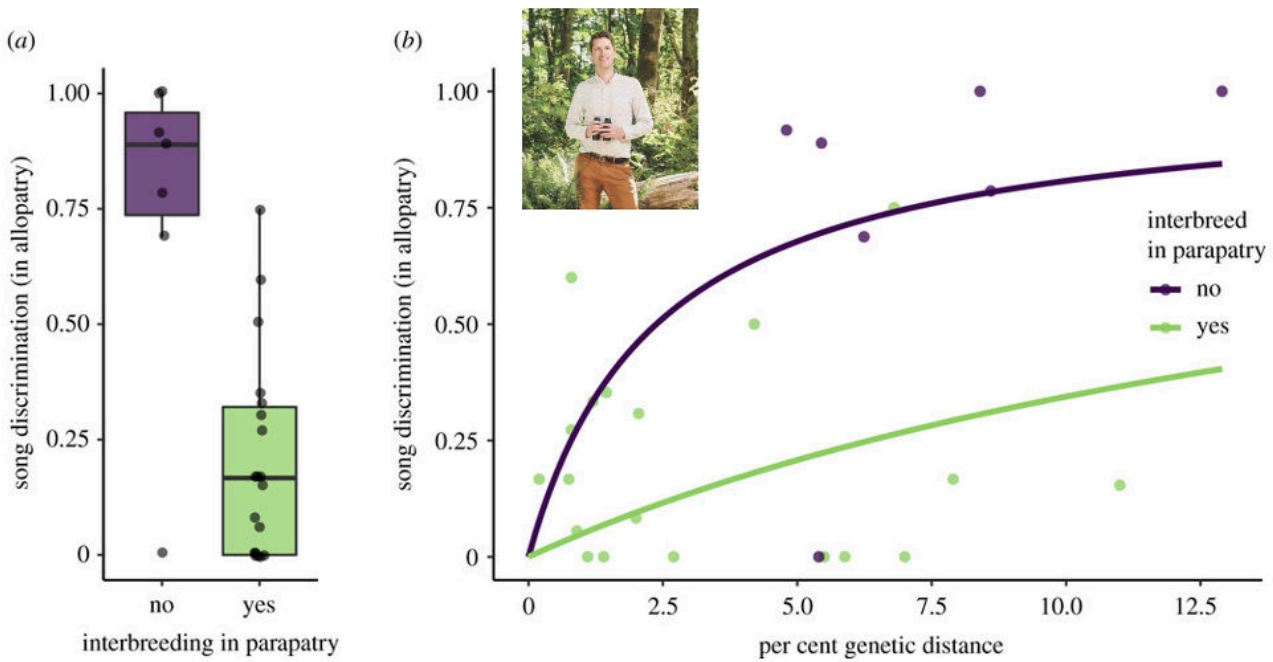
- Divergence *before* mating: ecology, behavior, appearance, song
- Can be measured with field experiments to test for intraspecific aggression against potential competitors for mates



(Freeman et al. 2022)

Prezygotic Isolation

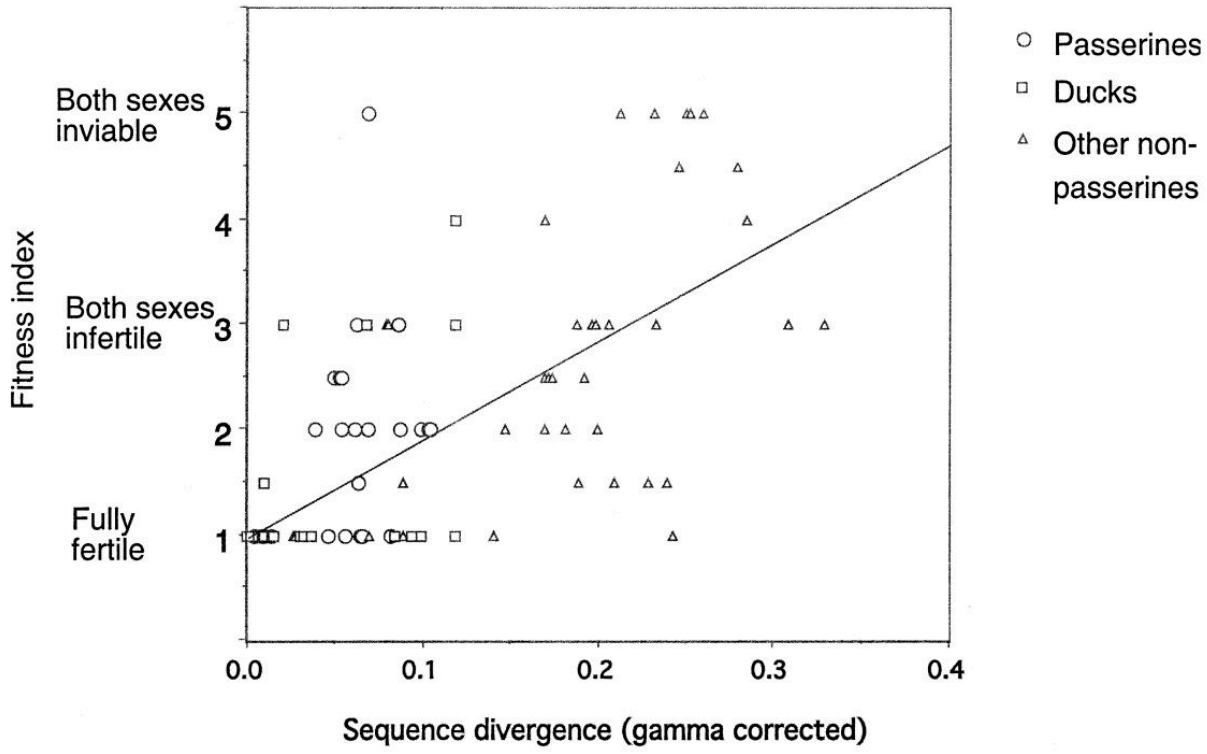
- Freeman et al. 2020: birds that do *not* respond aggressively to songs from close relatives ("song discrimination") in allopatry (isolation) are less likely to interbreed in parapatry (i.e., contact)
- Birds with the same level of genetic divergence but greater song discrimintation are less likely to interbreed in parapatry



(Freeman et al. 2022)

Postzygotic Isolation

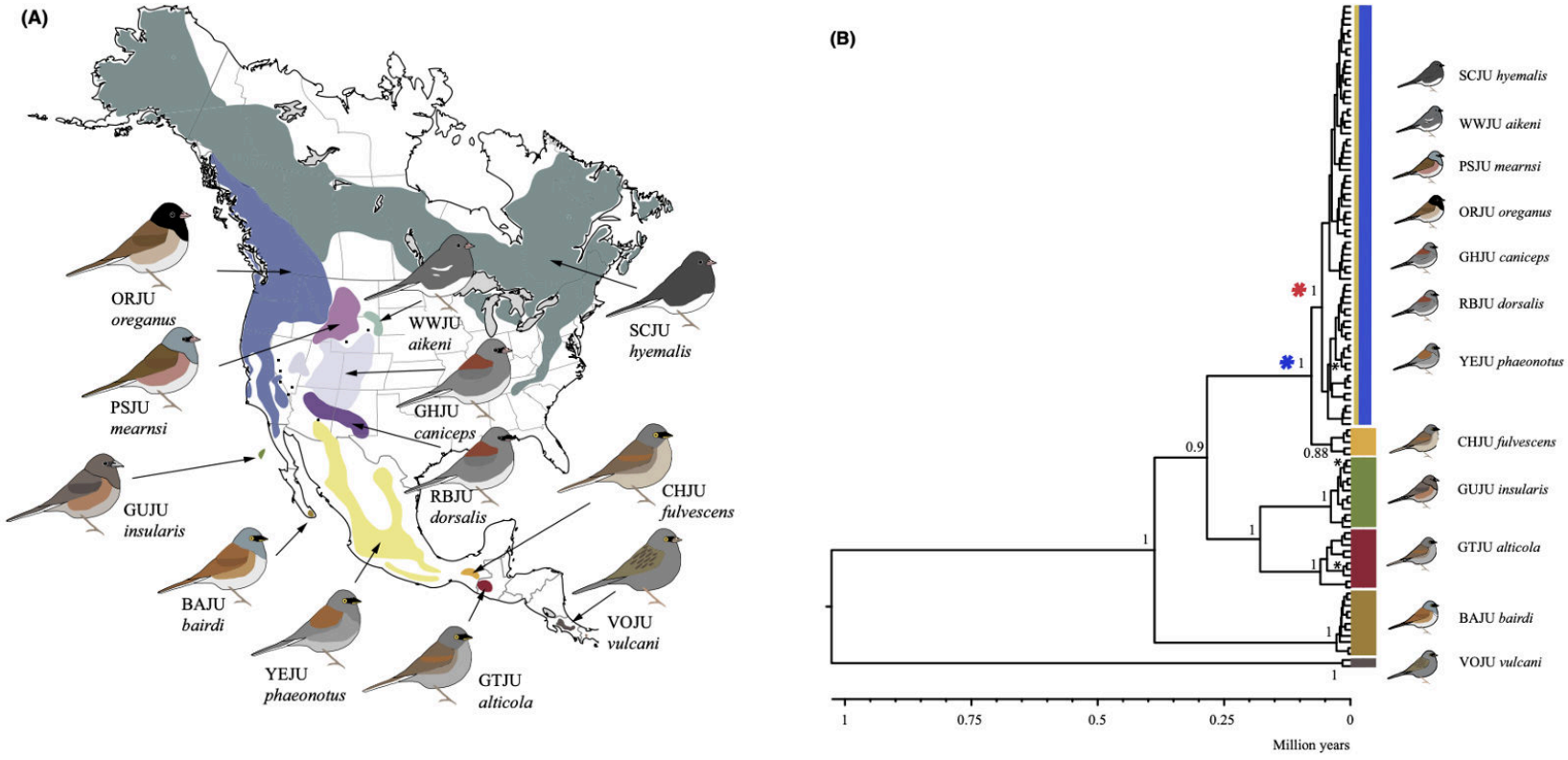
- Divergence *after* mating: genetic incompatabilities, selection against hybrids
- Can be measured by fitness or prevalance of hybrids, experiments, assessment of genetic architecture



(Price et al. 2002)

Allopatric speciation

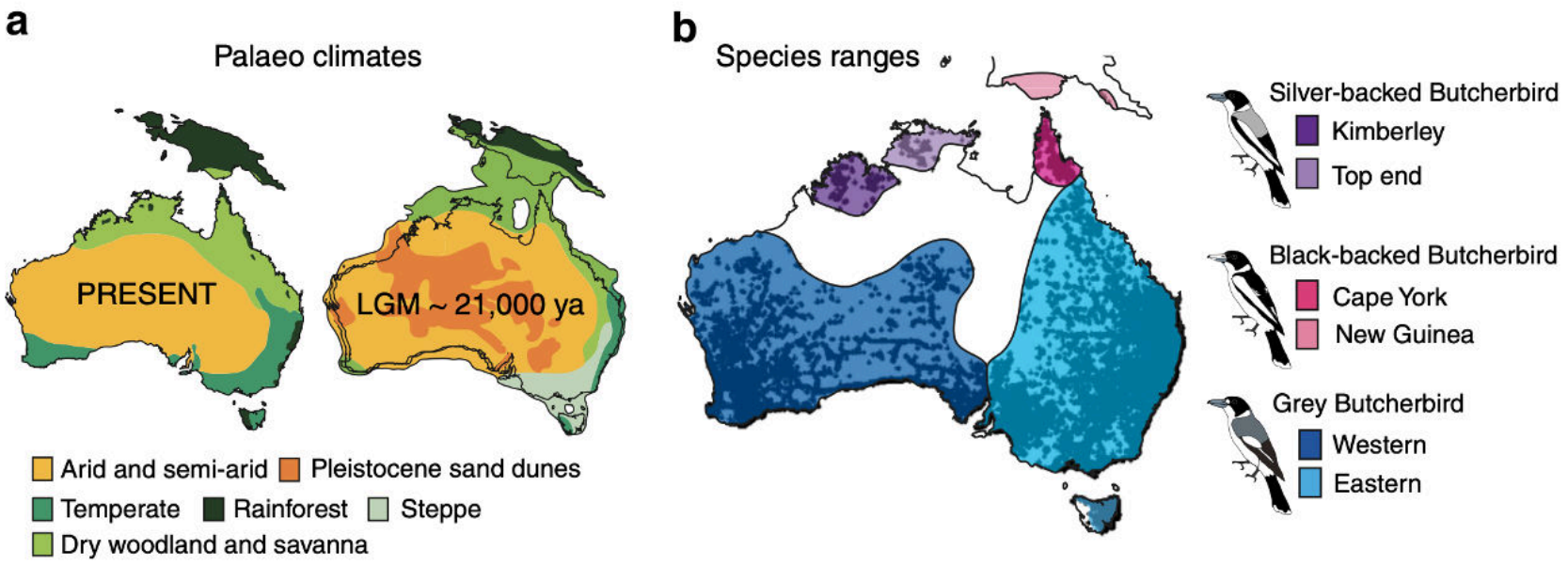
- Speciation in geographic isolation ("allo"=different, "patry"=homeland)
- Predicts allopatric populations are monophyletic groups



(Freis et al. 2016)

Allopatric speciation

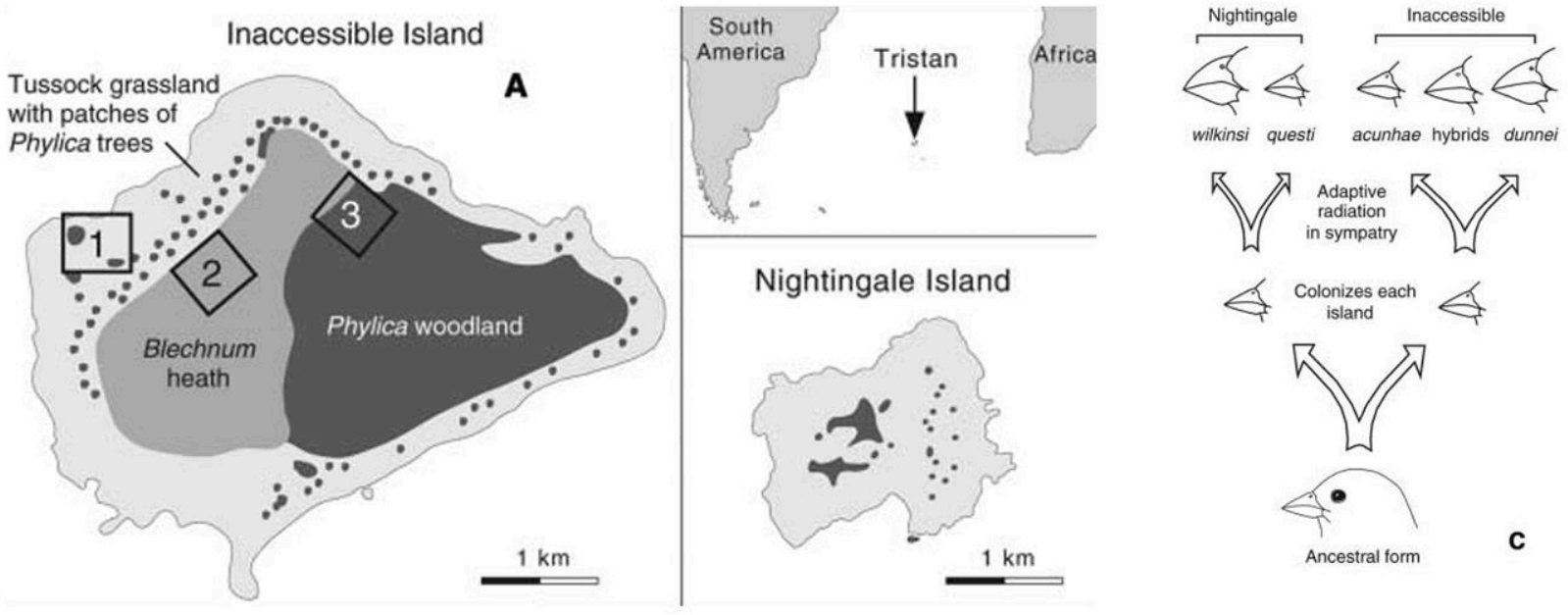
- "Easiest" and by far most common mode
- Allopatric ranges can be generated by geologic processes, climatic changes, dispersal



(Kearns et al. 2014)

Is sympatric speciation possible?

- Speciation *without* geographic barriers to gene flow
- Requires strong natural selection

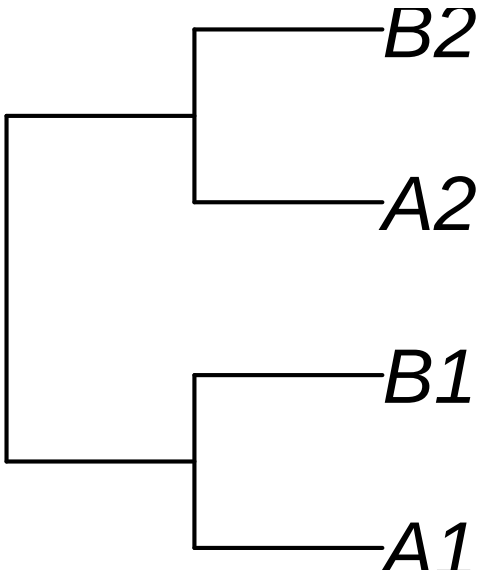


(Ryan et al. 2016)

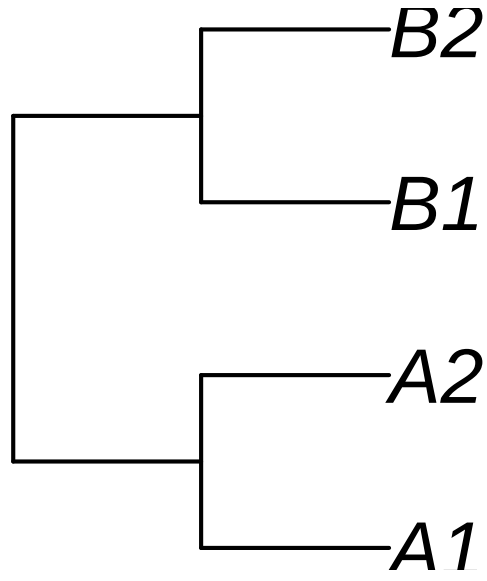
Which prediction is consistent with sympatric speciation?

- Tips with the same number are on the same island
- Tips with the same letter have the same ecological adaptation

a)



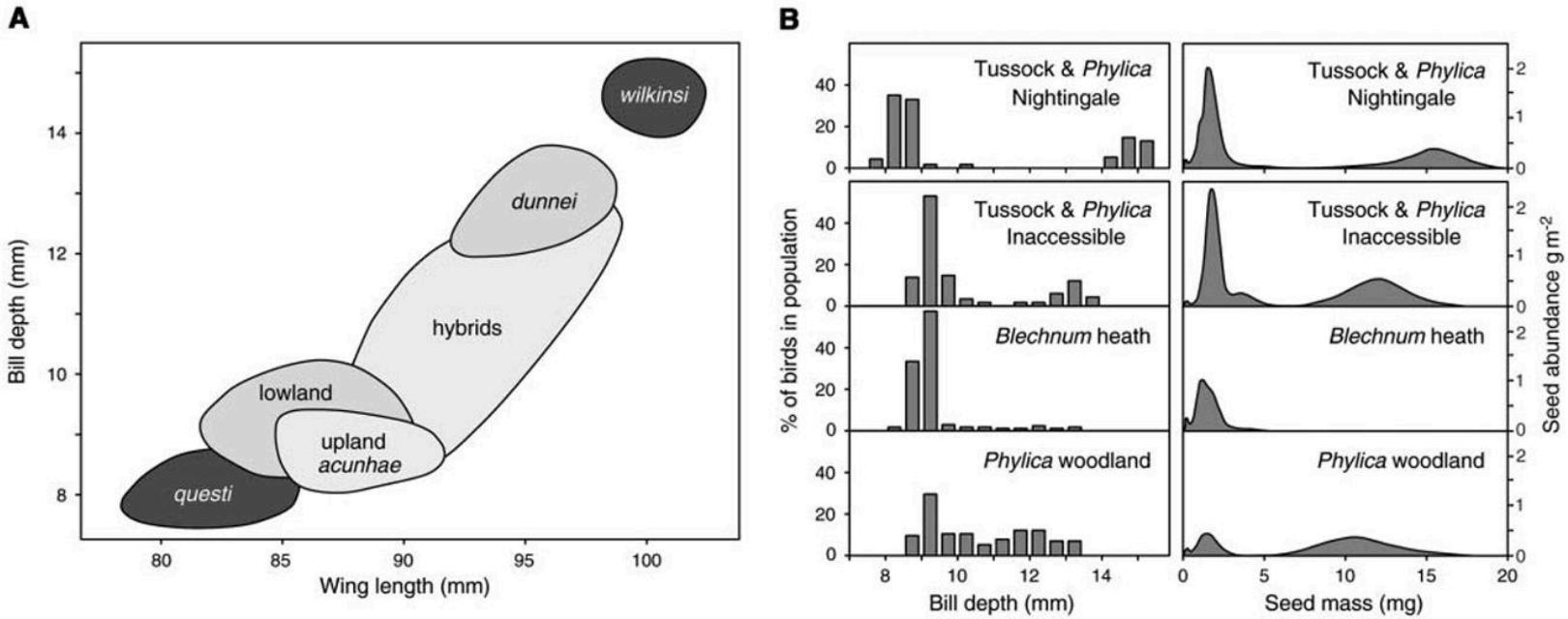
b)



(in **a**, birds on the same island have different niches; in **b**, birds on different islands have the same niche)

Sympatric speciation in South Atlantic Finches

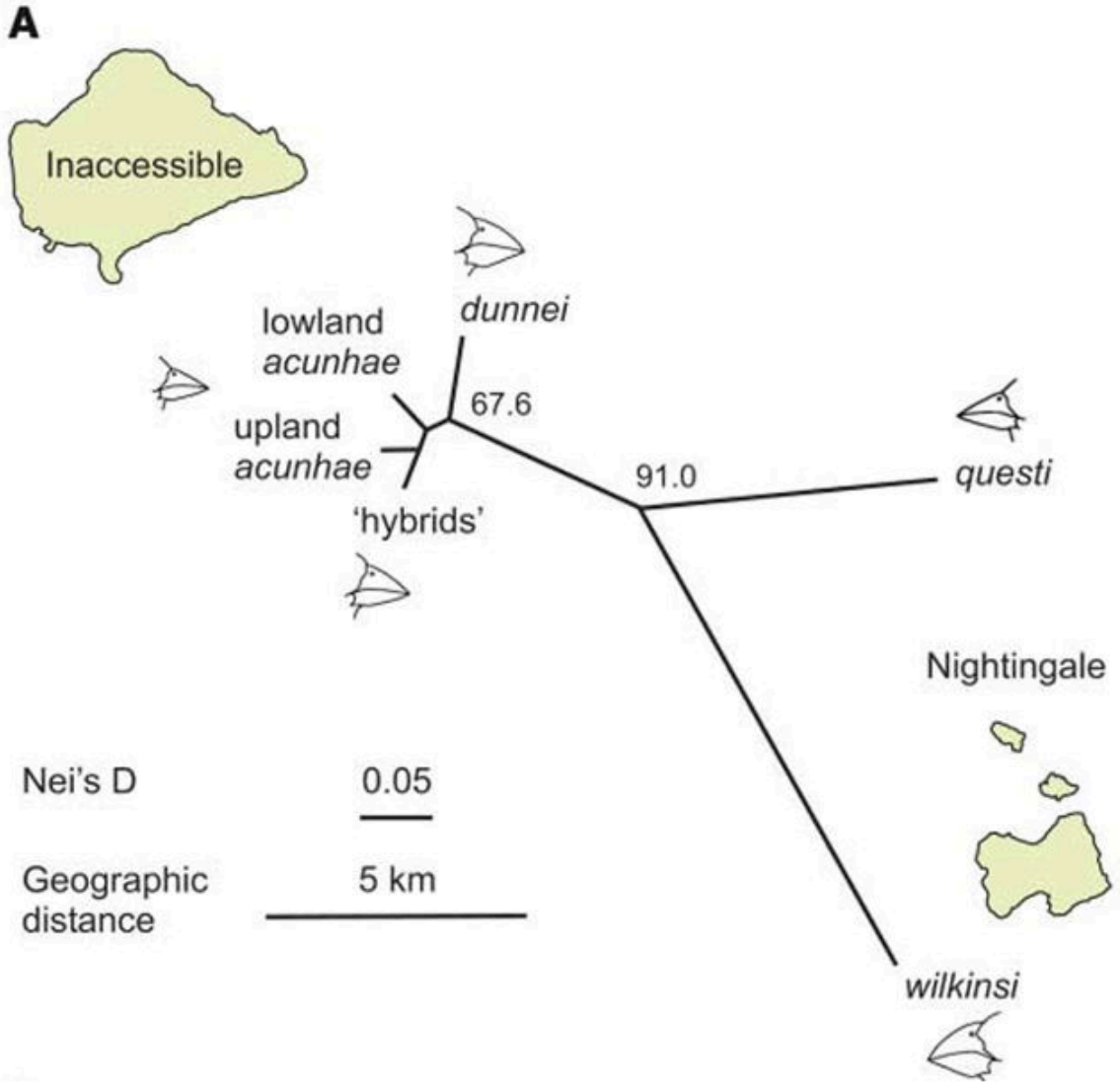
- Bill size matches seeds in habitat



(Ryan et al. 2016)

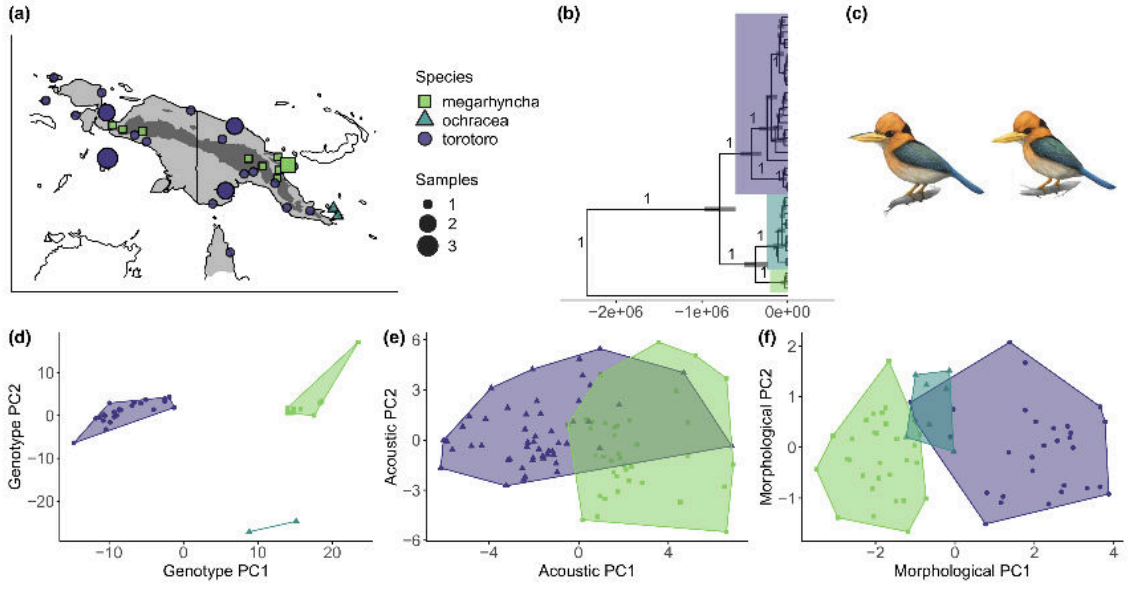
Sympatric speciation in South Atlantic Finches

- Different ecological forms within each island are most closely related



Parapatric speciation

- Intermediate to allopatric and sympatric speciation
- Adjacent but nonoverlapping ranges

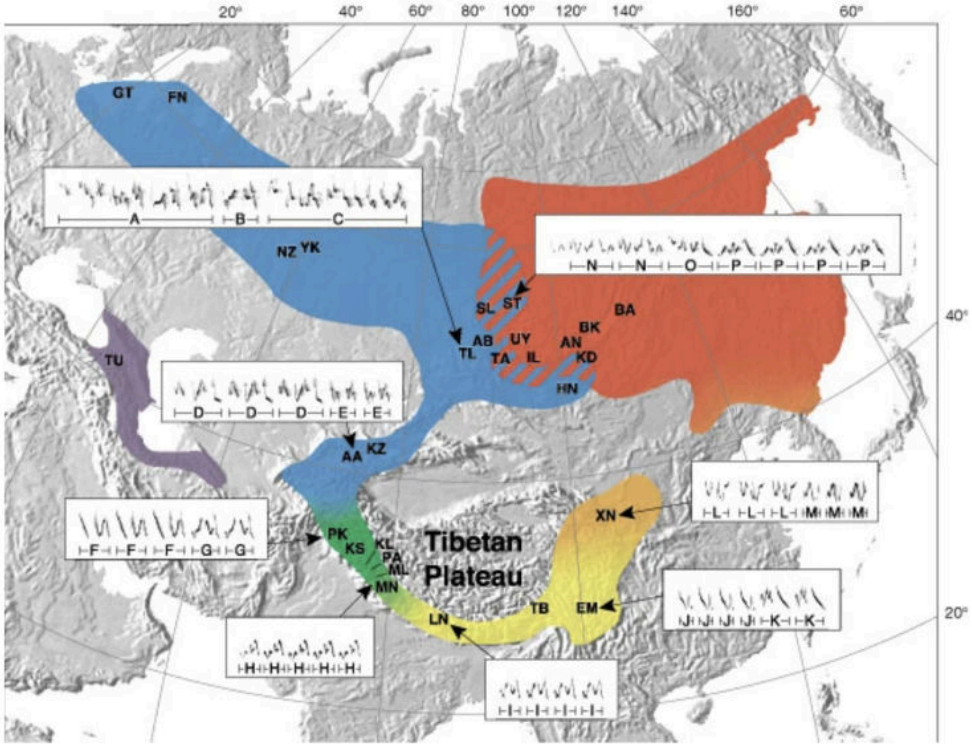


(Linck et al. 2020, *J. Evol. Biol.*)

(in Linck et al., Mountain Kingfisher diverged from Yellow-billed Kingfisher once, not repeatedly across elevational gradients, making paraptric speciation hard to prove, though they do hybridize)

Ring species

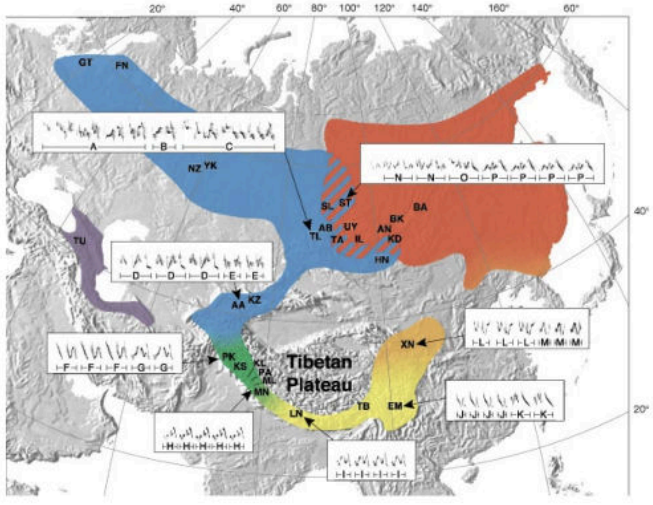
- Adjacent populations similar but subtly different
- Reproductive isolation increases with distance, complete at ends of horseshoe
- Example: Greenish Warbler wrapping around the Tibetan Plateau; morphologically and ecologically similar.





(Irwin et al. 2001; *Phylloscopus trochiloides*)

Would the Biological Species Concept split Greenish Warbler?





(Irwin et al. 2001; *Phylloscopus trochiloides*)

a) Yes
b) No
c) It depends on geographic sampling

Lecture Exam Review

- Wednesday, September 3rd
- 50 points on avian diversity, taxonomy, biogeography, phylogenetics, and speciation
- No lab content that wasn't also covered in lecture (e.g., what a hallux is)
- Question types: multiple choice, multiple select, true / false, fill-in-the-blanks, short answer
- Format: Canvas quiz, closed book. Bring a laptop or tablet.
- **Anything covered in lecture is fair game**

Diversity

- Know what an apomorphy is
- Know what traits define birds
- Be able to qualitatively describe how the number of described species has changed through time, and why
- Know what an adaptive radiation is
- Know how bird bodies, feet, bills, habitat use reflects the process of adaptive radiation

Taxonomy & Biogeography

- What taxonomic class do birds belong to?
- Know how many avian classes, orders, families, genera, and species there are to an order of magnitude
- Be able to qualitatively describe how the number of described species has changed through time, and why
- What latin suffix does each avian order have? Each family?
- Know at least one taxon (order, family, genus, or species) from each higher level group in the `03_taxonomy_biogeography.pdf` lecture
- Know at least one taxon associated with each biogeographic realm
- Know what "endemism" means, and what causes it

Evolutionary origins

- Why are birds considered reptiles in a phylogenetic sense?
- What traits do birds share with reptiles?
- Did hollow bones and feathers follow the evolution of flight, or vice versa?
- Modern bird traits present and absent in *Archaeopteryx*
- Bird-hipped versus reptile-hipped dinosaurs
- Cursorial versus arboreal hypotheses for the origin of flight
- K-Pg boundary, Mesozoic versus Cenozoic
- Cladogenesis versus anagenesis

Phylogenetics and Systematics

- Terminology: monophyly, paraphyly, apomorphy, polytomy
- Skills: Reading trees, identifying monophyletic groups
- Knowledge: describe / name members of Australaves, Paleognathae, Neoaves, Galliformes, Anseriformes, Aequiornithes, "Basal" Neoaves
- Remember: how taxonomy and phylogenetics are related, why relationships basal Neoaves are controversial
- What are the limitations of morphological data, and what are the advantages of DNA?
- Why do recent higher-level avian phylogenies conflict with each other?
- Why might two species be "lumped"?

Species concepts and speciation

- Be able to define / identify major species concept categories (e.g., phylogenetic versus biological versus ecological)
- Prezygotic versus postzygotic isolation (description and examples)
- Allopatric versus sympatric verys and parapatric speciation
- Which mode is most common, and how do we know?
- What is a ring species?

References:

- Title Slide Species: *Trogon citreolus*, *Pitta sordida*, *Icterus icterus*, *Pachycephala macrorhyncha*
- Kearns, A. M., Joseph, L., Toon, A., & Cook, L. G. (2014). Australia's arid-adapted butcherbirds experienced range expansions during Pleistocene glacial maxima. Nature communications, 5(1), 3994.
- Freeman, B. G., Rolland, J., Montgomery, G. A., & Schluter, D. (2022). Faster evolution of a premating reproductive barrier is not associated with faster speciation rates in New World passerine birds. Proceedings of the Royal Society B, 289(1966), 20211514.
- Price, T. D., & Bouvier, M. M. (2002). The evolution of F1 postzygotic incompatibilities in birds. Evolution, 56(10), 2083-2089.
- Irwin, D. E., Bensch, S., & Price, T. D. (2001). Speciation in a ring. Nature, 409(6818), 333-337.

References:

- Linck, E., Freeman, B. G., & Dumbacher, J. P. (2020). Speciation and gene flow across an elevational gradient in New Guinea kingfishers. Journal of Evolutionary Biology, 33(11), 1643-1652.
- Ryan, P. G., Bloomer, P., Moloney, C. L., Grant, T. J., & Delport, W. (2007). Ecological speciation in South Atlantic island finches. Science, 315(5817), 1420-1423.